Sandeep Kumar

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Education

Indian Institute of Technology Roorkee — Uttarakhand, India
Integrated Dual Degree (Masters+Bachelors) in Applied Mathematics — *2020 – 2025*

Experience

Standard Chartered Bank — July 2025 – Present
Software Development Engineer | React.js, Python Flask, PySpark, REST APIs, Oracle — *Bengaluru*

- Architected a full-stack transaction automation platform (**PT HARNESS**) using React.js, Python Flask, and REST APIs, automating end-to-end enterprise payment processing workflows and reducing manual intervention by **45%**.
- Engineered high-throughput ETL **data pipelines** with PySpark and Oracle DB for cross-system transaction reconciliation and real-time SLA monitoring across hybrid OLTP/OLAP environments.
- Automated asynchronous report generation and execution lifecycle tracking across **20+** enterprise modules, improving test execution visibility and cutting reporting turnaround time by **70%**.
- Containerized and deployed microservices on virtualized Linux infrastructure, leveraging Git CI/CD workflows and modular Python package architecture for low-latency production delivery.

Standard Chartered Bank — May 2024 – July 2024
Software Development Intern | React.js, Spring Boot, PostgreSQL, Large Data Handling — *Chennai*

- Engineered a role-based case tracking dashboard using React.js and Spring Boot to streamline priority, pending, and overdue workflows, enhancing operational tracking efficiency for **1,000+** daily enterprise operations.
- Developed an interactive visual query builder over PostgreSQL enabling non-technical users to generate complex SQL joins, aggregations, and data visualizations, cutting manual query construction time by **40%**.

WorldQuant Brain — June 2023 – April 2024
Research Consultant | Financial Analysis, Mathematical Modeling — *Remote*

- Formulated and backtested **50+** equity long–short alpha strategies using statistical arbitrage and factor-based modeling, improving signal selection through risk-adjusted performance filtering.
- Automated alpha generation and **backtesting** on ACE in Python, filtering strategies via risk and performance metrics.

Projects

Connect 4 AI Agent — Aug 2024 – Nov 2024

- Modeled the 6×7 grid as a finite **Markov Decision Process** (MDP), leveraging state space S, action space A, and reward R to evaluate optimal move-state transitions.
- Designed **Deep Q-Learning (DQN)** for non-linear state-action value approximation achieving a **99.38%** win rate over $\mathbf{4.5 \times 10^{12}}$ game states.
- Minimized **Temporal Difference** (TD) error δ_i via **Huber Loss**, updating network parameters against **Bellman Target**.
- Formulated a linear output layer predicting **7 action Q-values**, using an **exploration-exploitation** policy to drive convergence.

Physics-Informed Neural Networks Solution for Lid-Driven Cavity Flow — Jan 2025 – April 2025

- Solved 2D incompressible **Navier–Stokes equations** for lid-driven cavity flow using a custom Physics-Informed Neural Network (PINN) architecture without relying on external simulation datasets.
- Embedded continuity and non-linear momentum equations into the loss function via **automatic differentiation**, training over **10,000+ collocation points** to minimize PDE residual loss to $< \mathbf{10^{-4}}$.
- Evaluated flow behavior across Reynolds numbers **Re = 100–1000**, achieving $< 1.5\%$ **relative** L_2 **error** against the **Ghia et al. CFD benchmark** for centerline velocity profiles and primary vortex locations.

Skills

Languages: Java, Python, C++, SQL
Backend Frameworks & APIs: Spring Boot, Flask, RESTful APIs
Distributed Systems & Architecture: System Design (HLD & LLD), Microservices, High Availability, Load Balancing, Fault Tolerance
Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, RabbitMQ, AWS SQS
Cloud, DevOps & Infrastructure: Docker, Kubernetes, AWS (EC2, S3, RDS), Linux/Unix, CI/CD, Grafana

Achievements

Codeforces (rating - **1802**, handle: Sandeep5445), Codechef (rating - **2060** , handle: sandeep_1132).
Secured **141st** (2nd in India) position in ICPC 2023 Online Spring Challenge powered by Huawei.
National Finalist – International Quant Championship 2023, WorldQuant Brain.
Secured **Ranks 186 and 1030** in Meta Hackercup Round 3 (2025) and Round 2 (2023), respectively.
Secured **Ranks 17 and 37** in CodeChef Starters 241 and Starters 209, respectively.
Secured **Ranks 64 and 110** in Codeforces Round 867 and Round 1058, respectively.